

DARIO LAZZARA

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Education

- Università degli Studi di Catania** Current
M.S. in Computer Science: Artificial Intelligence and Machine Learning
 - Focus: Deep Learning, NLP, Cloud Computing, Social Media Data Analysis.
- Università degli Studi di Catania** Sep. 2025
B.S. in Computer Science: Artificial Intelligence & Robotics | Grade: 106/110
 - Thesis: Extracted key behavioral insights from Open Surgery procedures and achieved a 60% recognition accuracy for hands and instruments across 17+ hours of highly occluded egocentric video data. 📺 📱
- Università degli Studi di Catania** Feb. 2023
B.S. in Political Science | Grade: 108/110
 - Thesis: Applied Neural Networks to 58k survey responses for social data analysis to forecast migration trends with 85.4% accuracy.

Technical Skills

Languages & Frameworks: Python, C/C++, SQL, Bash, Pytest
Artificial Intelligence (ML/NLP/CV): PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, GenAI
Data Engineering: Pandas, NumPy, Matplotlib, Playwright, Selenium, Regex
DevOps, Cloud & Tools: AWS, Kubernetes, Docker, Linux, Git, CI/CD, SOLID

Projects

- Semantic Alignment in Transformer Models** 🔑 2026
 - Fine-tuned RoBERTa with a Triplet strategy and MNRL to resolve "Context Domination" and improve emoji retrieval accuracy to 90.7% Top-5 (vs 1.7% baseline).
 - Engineered a custom pipeline using PyTorch for Unicode normalization and hard-negative mining to optimize semantic alignment across a 100M tweet corpus.
- Aspect-Based Sentiment Analysis in FinTech** 🔄 2026
 - Developed a Playwright scraper and a Gemini 2.0 Flash BIO-tagging pipeline to automate ground-truth labeling for 6,500+ banking reviews.
 - Fine-tuned an LCF-ATEPC architecture on an Italian BERT backbone to reach 87.6% sentiment classification accuracy and 61.6% aspect extraction F1-score.
- Blind Super-Resolution & Denoising via SRGAN** 🔄 2026
 - Adapted an SRGAN architecture for simultaneous 4x upscaling and denoising to boost image fidelity by +5 dB PSNR and raise SSIM to 0.74.
 - Implemented a two-stage training strategy (MSE-optimized SRResNet pre-training followed by Adversarial fine-tuning) to eliminate Gaussian noise and mosaic artifacts.
- BoardGuard: Edge AI & IOT Security** 🔄 📺 2025
 - Deployed a low-latency Edge AI model on ESP32 to detect boat theft using real-time motion and speed sensor data.
 - Designed a two-layer encryption protocol over a LoRaMesh network to secure decentralized long-range vessel communication.
- FlyNow (AI Travel Planner)** 🔄 2024
 - 2nd Place Winner, NeoData Hackathon. Led a team of 3 to build an AI prototype using OpenAI APIs to reduce user planning time by 60–70%.

Extracurricular Activity

- Quality Development** 🔄 Nov. 2025
 - Developed a Pacman clone in Python applying Object-Oriented Design (SOLID) and UNIX shell scripting.
 - Implemented unit tests with *pytest* and established CI/CD pipelines via GitHub Actions for automated deployment.
- Intelligent Agents and Machine Learning** July 2025
 - Completed 75 hours of advanced coursework on multi-agent architectures and their applications in AI.
 - Developed local multi-agent projects using LangChain, LangGraph, and Ollama for autonomous task execution.
- AI & Robotics with Micro-controllers** June 2024
 - Applied core concepts of TinyML and robotics for Edge computing and passed the final assessment with a 100% score.